Pixel-Based Optical Navigation
Near Earth-Moon L1

EKF Estimation,
Midcourse Correction,
and Feature Tracking

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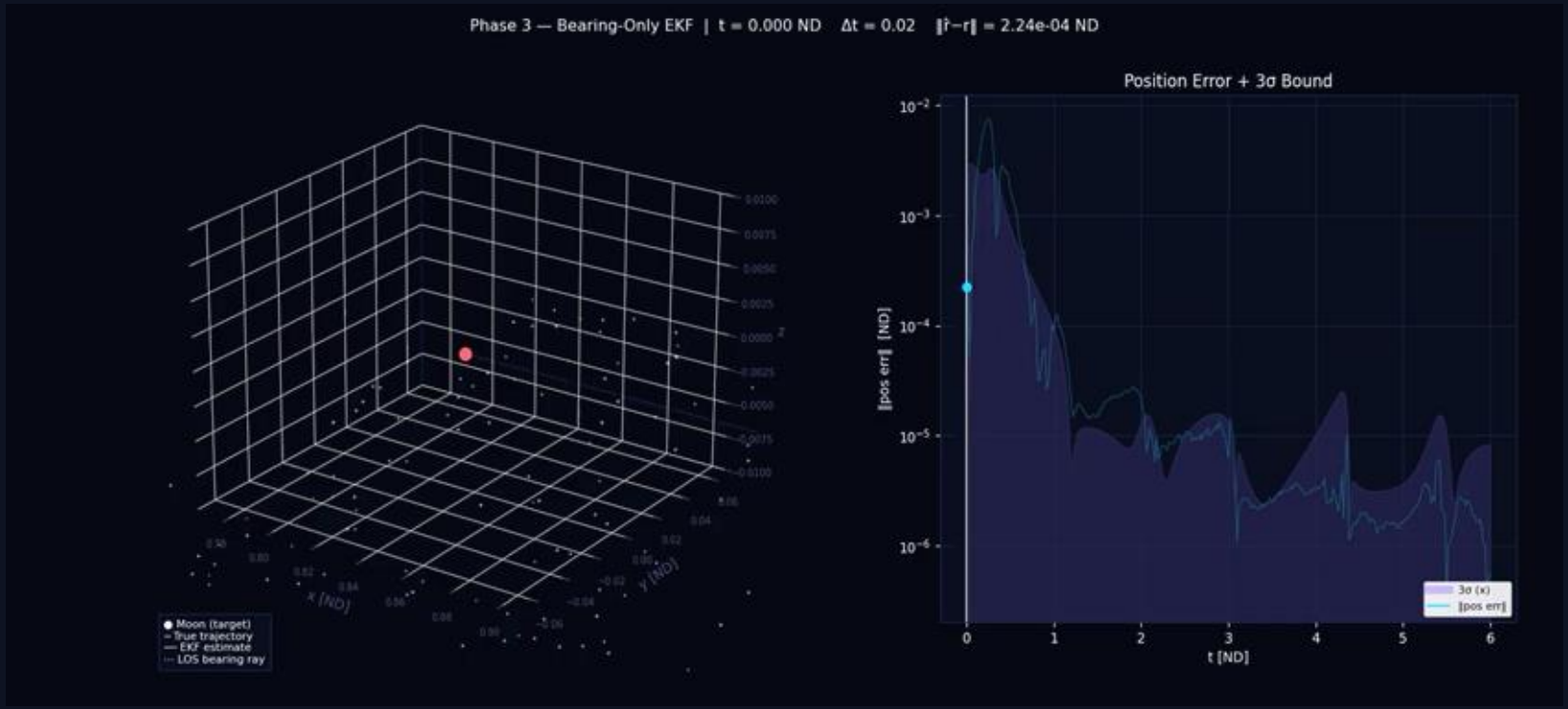
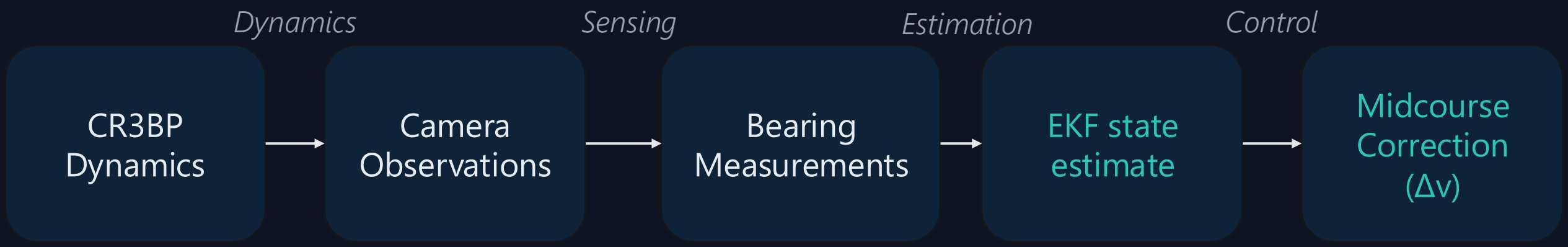
Problem: Autonomous Navigation in Cislunar Space

- Limited GPS and ground-based tracking in cislunar space
- Need for onboard, autonomous navigation
- Optical sensing (camera) provides line-of-sight measurements
- Pixel observations give no direct range information
- Observability depends on geometry and sensor pointing

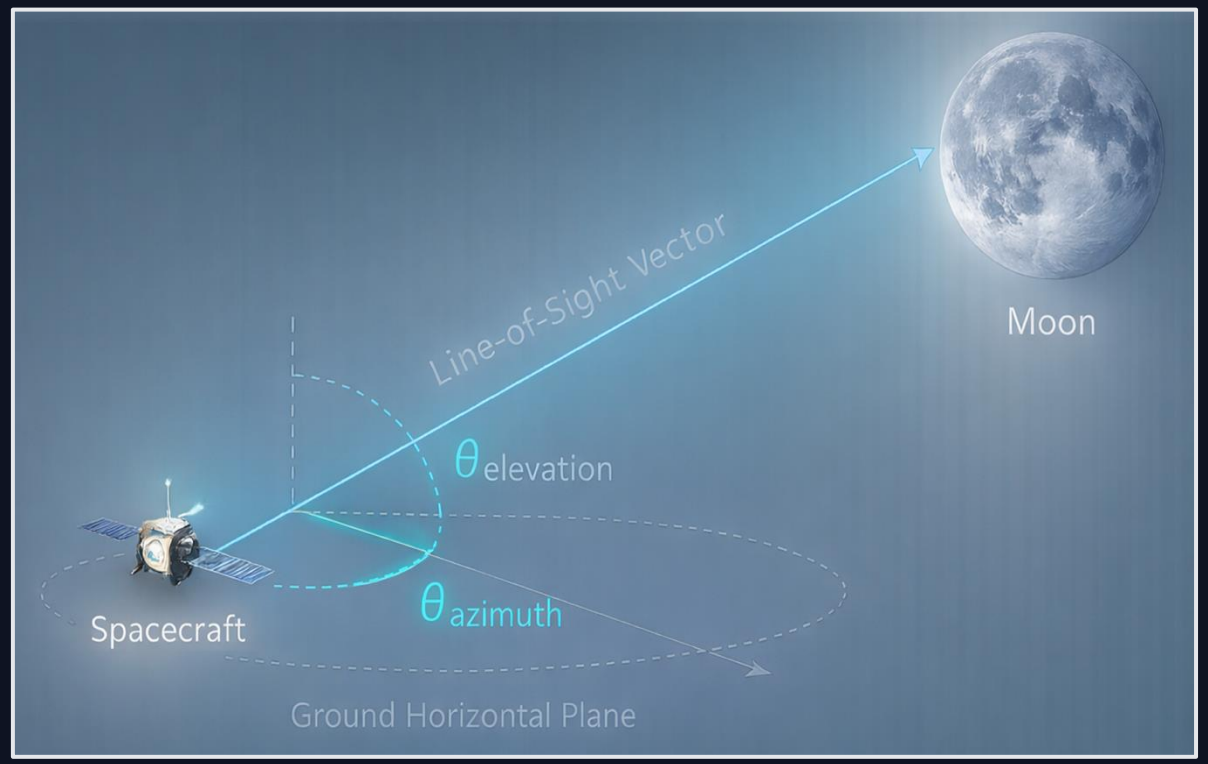


Optical navigation introduces a fundamentally bearing-only estimation problem

System Overview



Measurement Model: Bearings-Only Observations



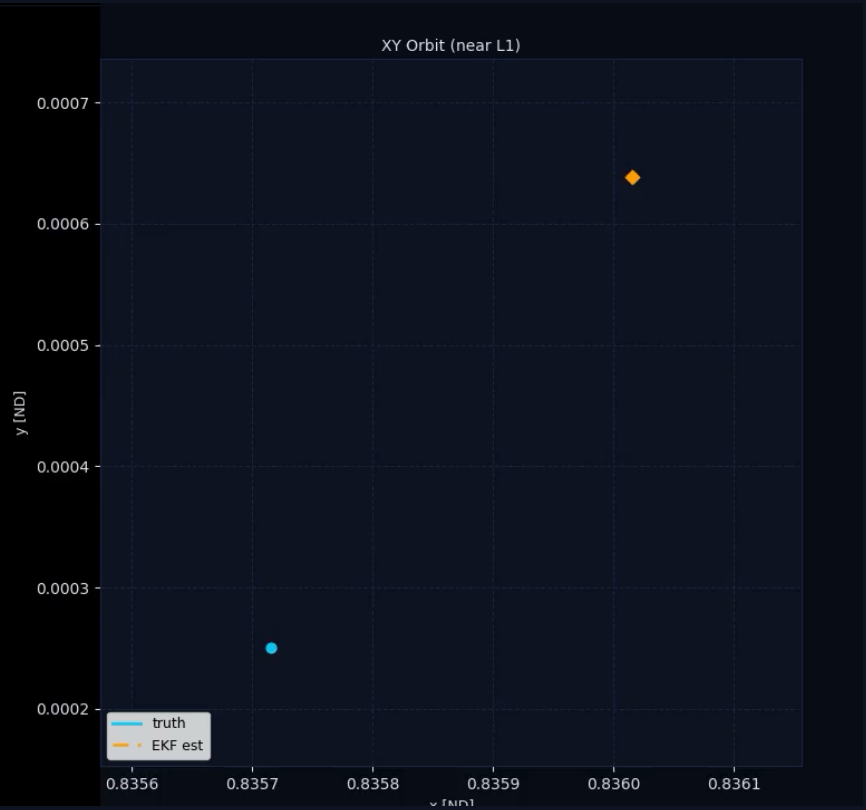
Camera → line-of-sight direction
Pixel measurements define a bearing (angle)
Measurement constrains direction only

Key limitation
No direct range information
Distance along LOS is unknown
Partially observable system

Range must be inferred from motion

Sensor Pipeline

Pixel --> Bearing --> EKF



Measurement generation

Bounding box detection

Centroid extraction → pixel (u, v)

Projection to line-of-sight bearing

Realistic sensing effects

Measurement noise & centroid bias

Intermittent detection / dropout

Gated updates (accept / reject)

State Estimation via Extended Kalman Filter

State Propagation (CR3BP)

↓

Predicted State

↓

Measurement Update (Bearings)

↓

Corrected State Estimate

Nonlinear state estimation

- Propagate state using CR3BP dynamics
- Incorporate bearing measurements

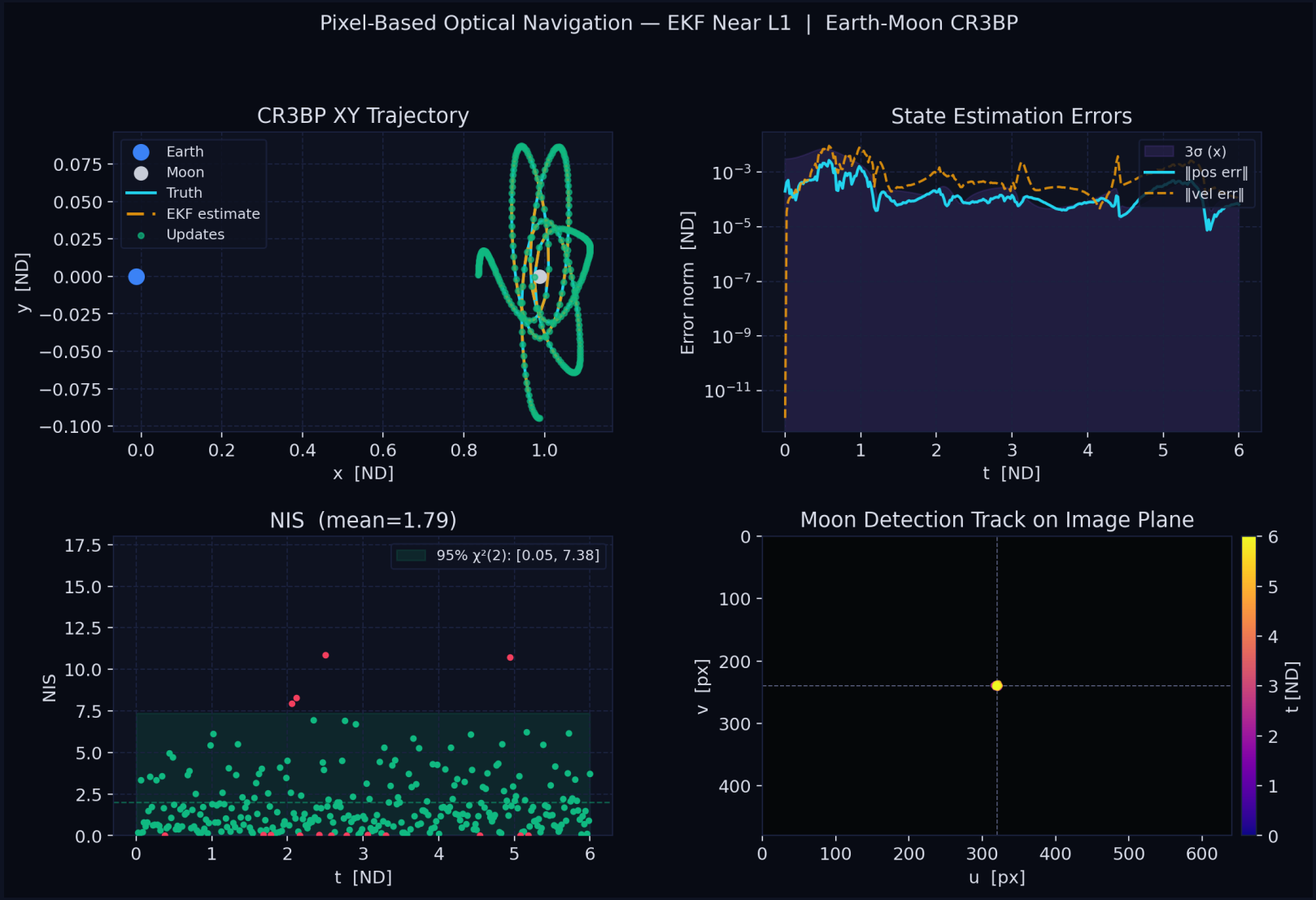
Key capability

- Combines dynamics + measurements
- Recovers full state from partial observations
- Tracks estimate uncertainty (covariance)

Consistency evaluated via NIS / NEES

Initial EKF Performance and Limitations

Initial EKF Performance and Limitations



Observed behavior

- Position error decreases but plateaus above ideal accuracy
- Estimated trajectory deviates slightly from truth

Statistical consistency

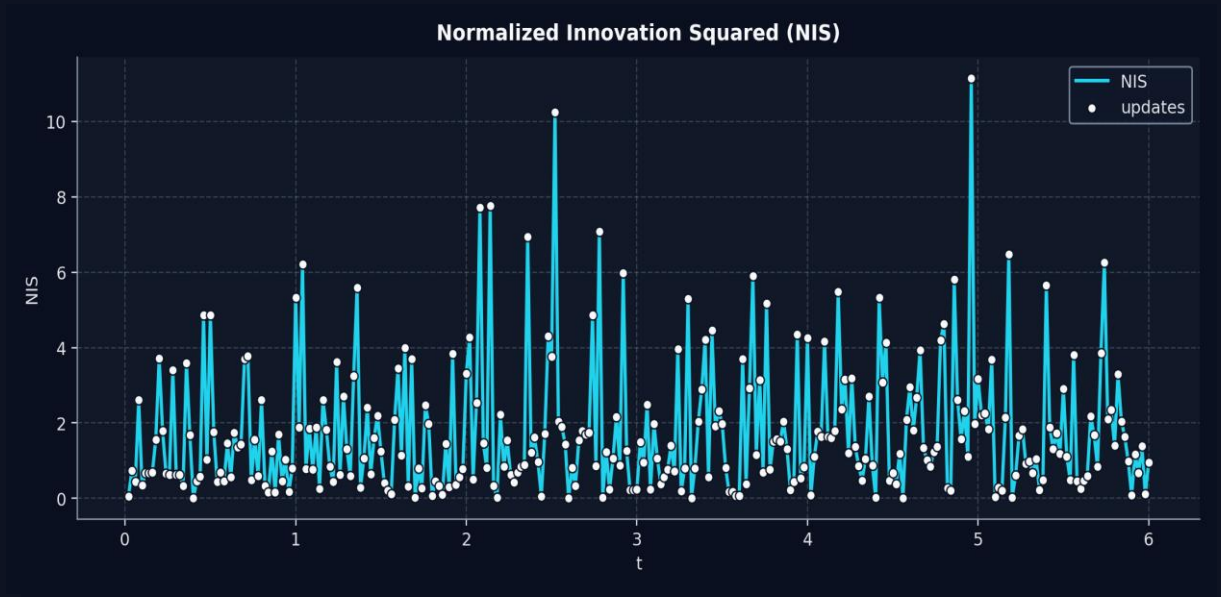
- NIS mostly within bounds with occasional outliers
- Indicates mild measurement-model mismatch

EKF performance is limited by measurement modeling and observability

EKF Validation and Diagnostics

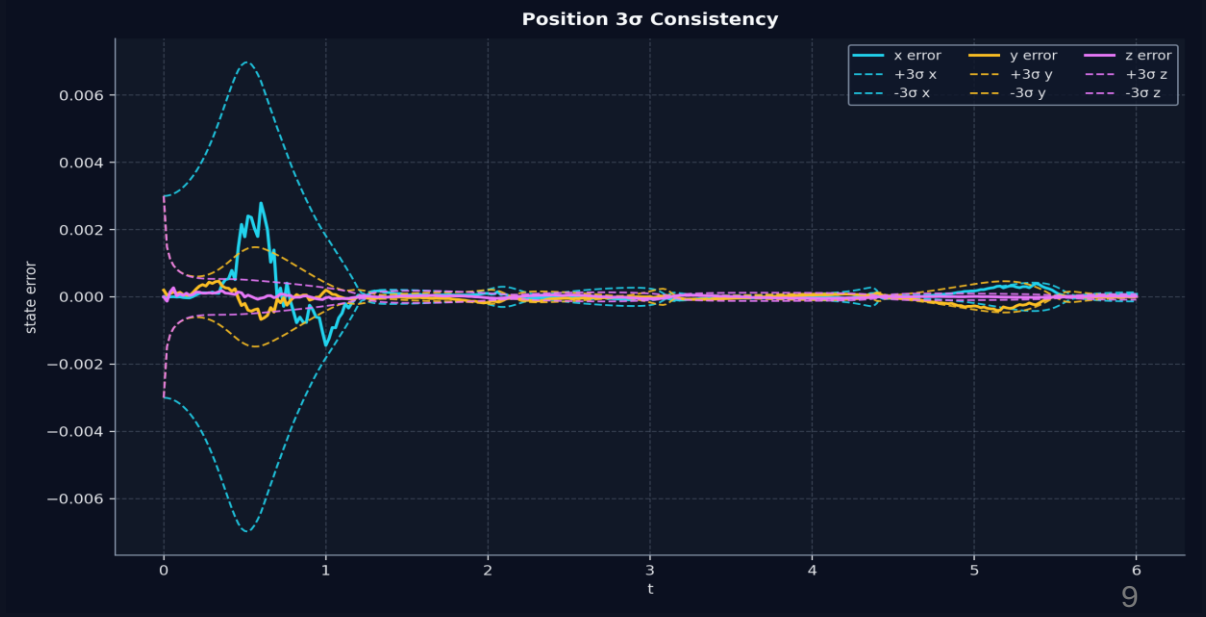
Validation approach

- Jacobian verified via finite differences
- State transition matrix (STM) validated
- NIS / consistency metrics evaluated

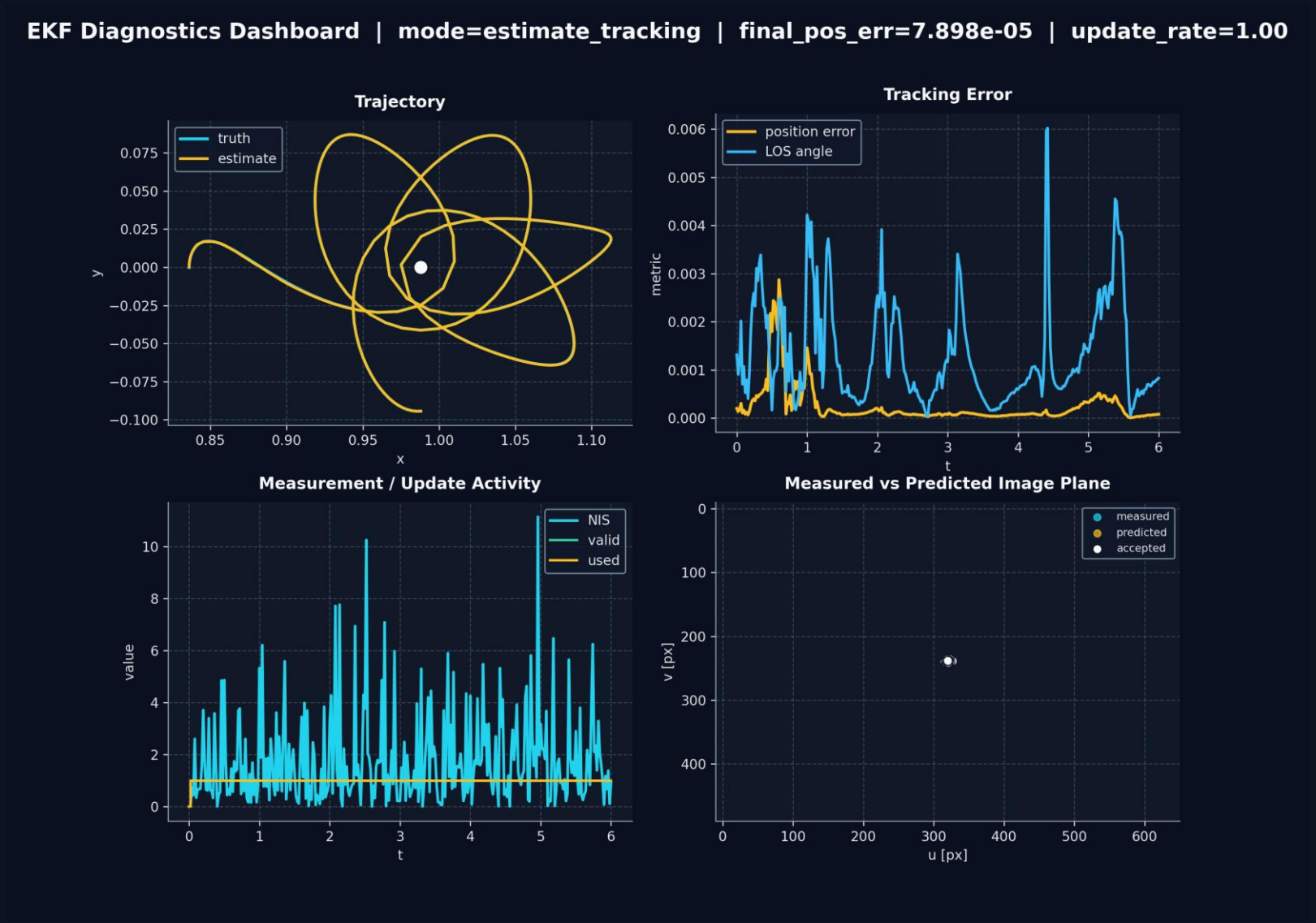


Key findings

- Bearing measurement model is accurate
- Filter is statistically consistent (NIS & 3σ bounds)
- Tracking camera mode provides continuous observability



EKF Validation and Diagnostics



Validated EKF achieves consistent, accurate state estimation under tracking geometry

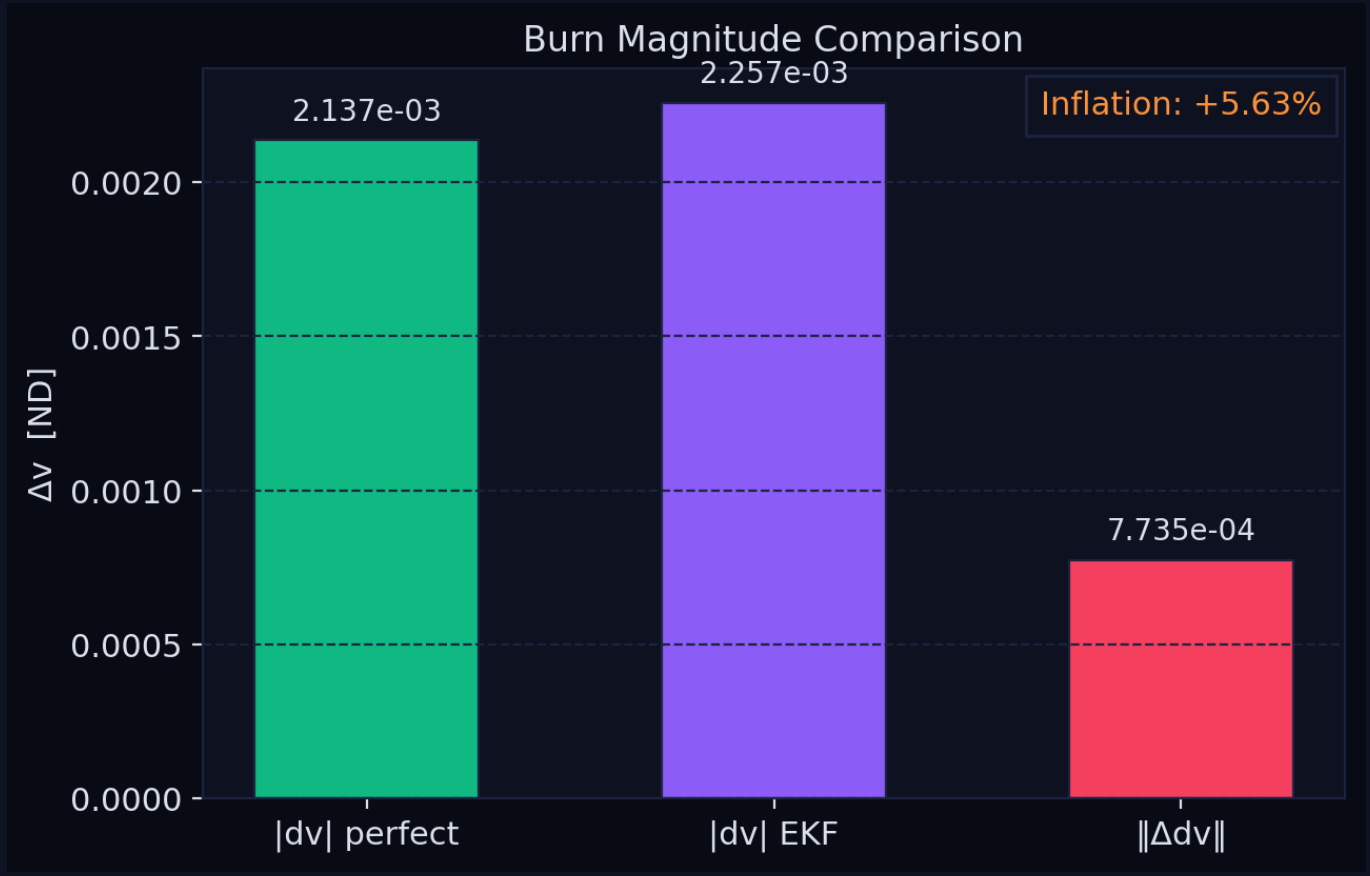
Navigation-Guided Midcourse Correction

Approach

- EKF state used for midcourse correction at t_c
- Δv computed to target L1 trajectory
- Compared against perfect-state solution

Results

- EKF-guided Δv within 5.6% of optimal
- Reduces miss distance by >60% vs uncorrected
- Achieves accurate targeting using onboard estimation only



Navigation-Guided Midcourse Correction

Autonomous optical navigation enables near-optimal trajectory correction without ground truth

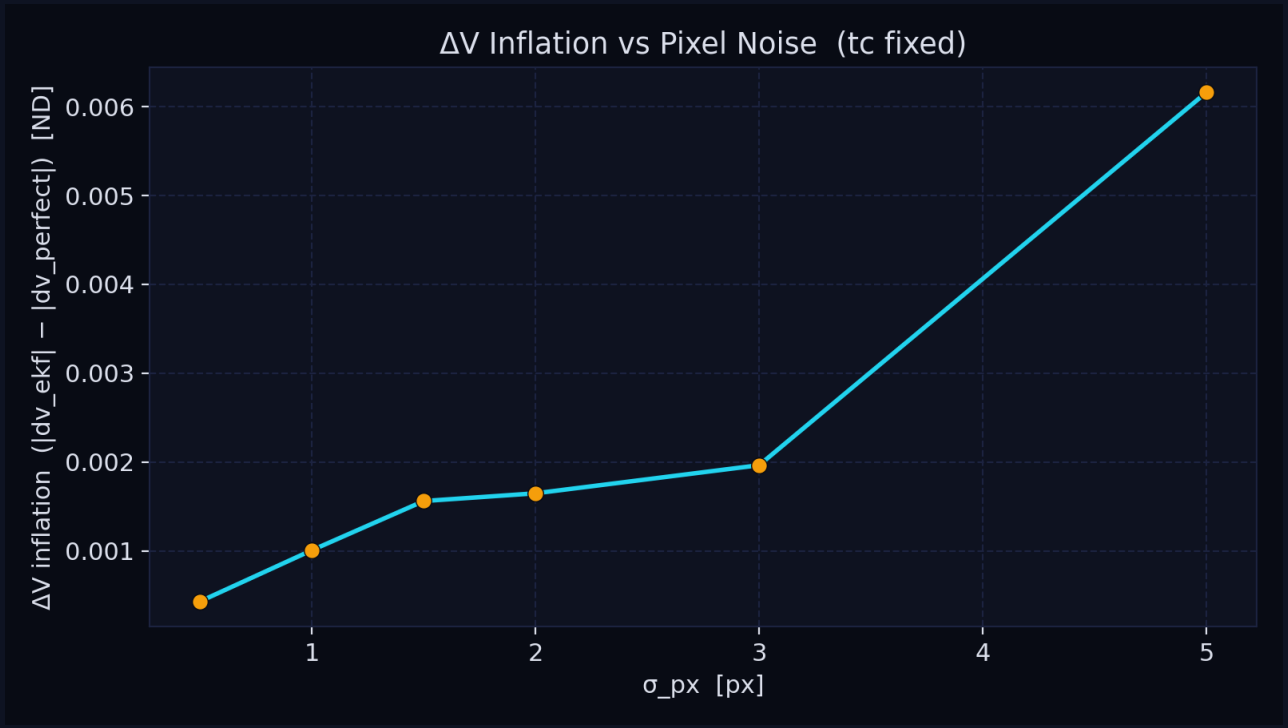
Sensitivity to Measurement Noise

Setup

- Pixel noise σ_{px} varied from 0.5 → 5 px
- Correction time t_c fixed
- Tracking camera mode

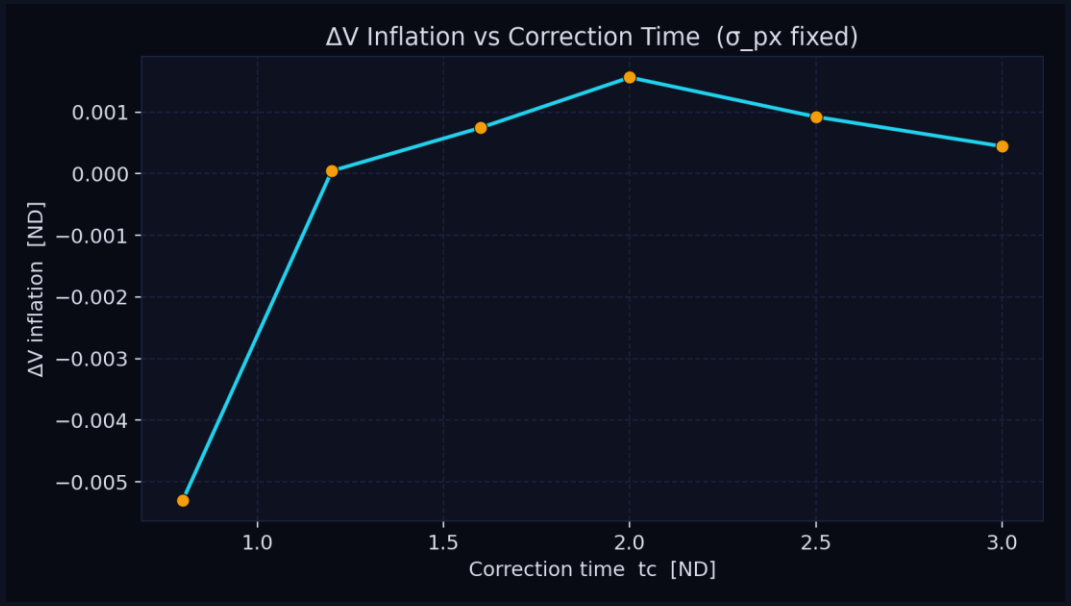
Key observation

- Δv inflation increases with measurement noise
- Low-noise regime → near-optimal performance
- High noise degrades targeting accuracy



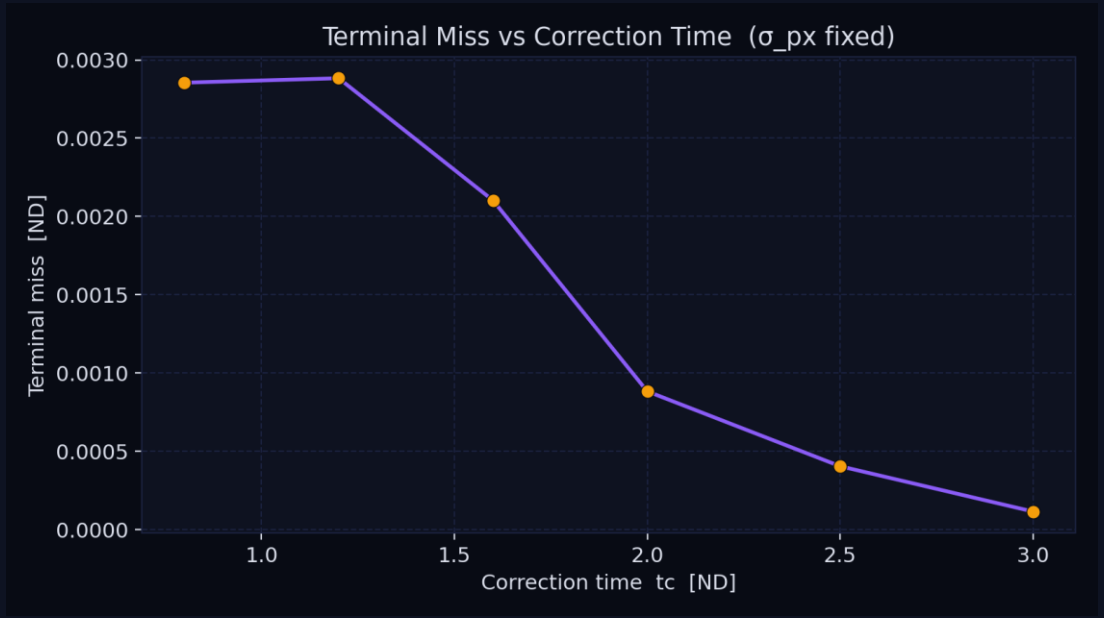
Navigation performance is robust at low noise, but degrades gracefully with measurement uncertainty

Sensitivity to Correction Timing



Setup

- Correction time t_c varied along trajectory
- Pixel noise fixed
- EKF-based state used

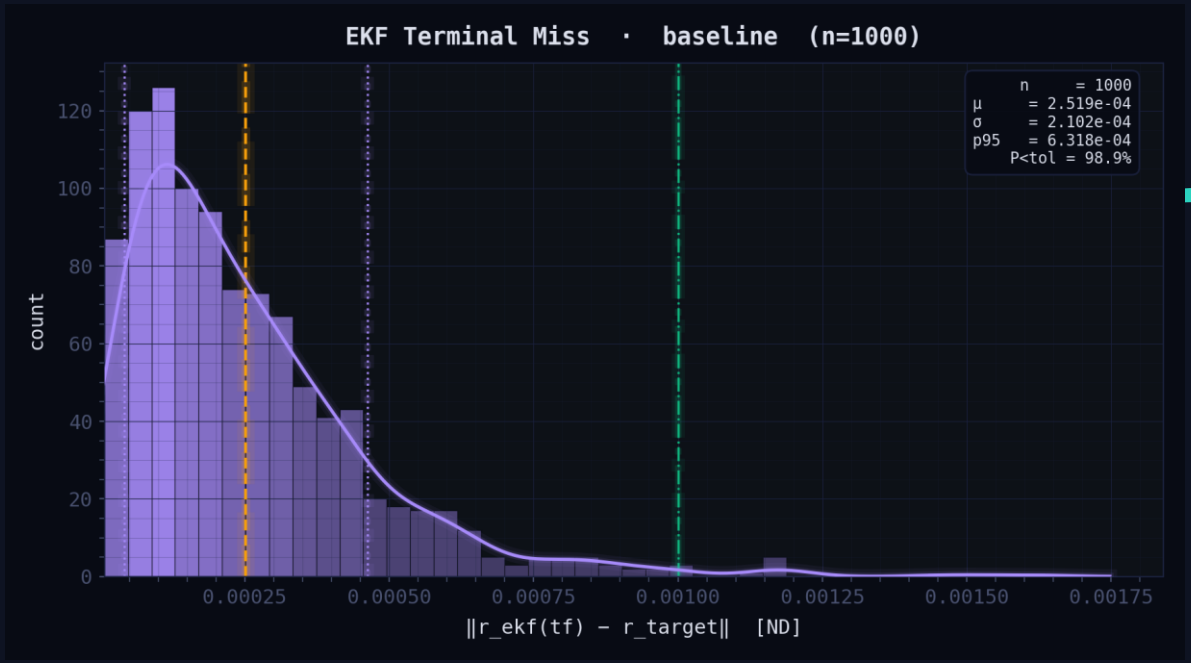


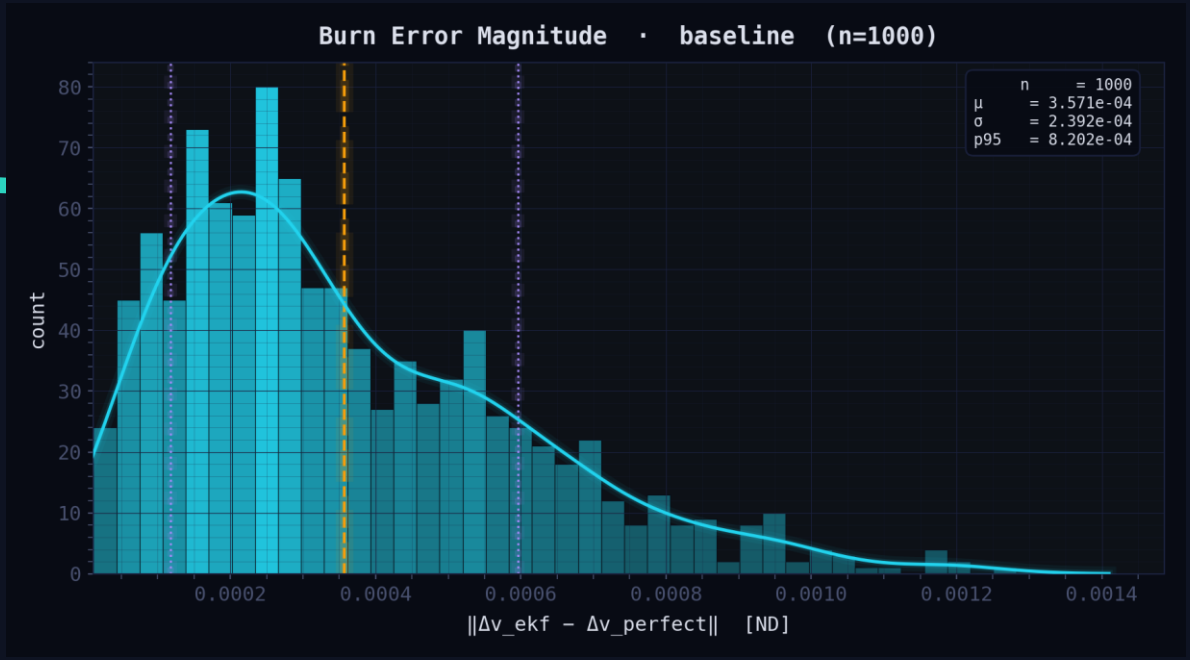
Key observation

- Early corrections are inefficient (high Δv inflation)
- Later corrections reduce miss distance significantly
- Optimal window exists for correction timing

Correction timing strongly influences efficiency and accuracy — later corrections improve targeting

Monte Carlo Robustness





1000 Monte Carlo trials

Randomized:

- Initial state errors
- Measurement noise
- Detection variability

Burn error tightly distributed

- Mean ≈ 3.6e-4 ND
- 95% < 8.2e-4 ND

Terminal miss remains small

- Mean ≈ 2.5e-4 ND
- 98.9% within tolerance

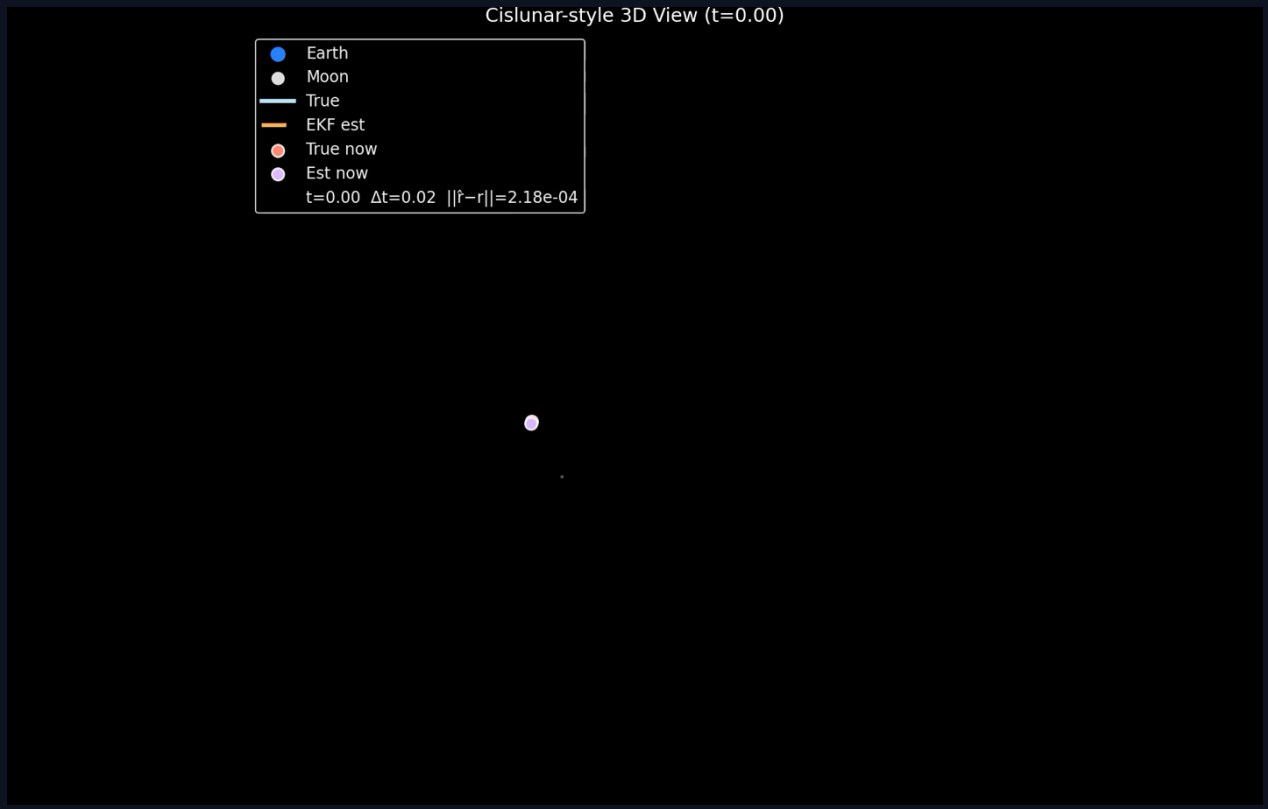
The navigation + guidance pipeline is statistically robust under realistic uncertainty

Key Conclusions

Navigation
Pixel-based optical navigation enables GPS-independent state estimation
Bearing-only measurements sufficient with proper geometry & tracking

Estimation
EKF achieves accurate and consistent convergence
Validated via:
- NIS consistency
- Jacobian / STM checks
- Diagnostics suite

Guidance
Navigation-driven midcourse correction:
- Reduces terminal error by >2×
- Maintains low ΔV inflation (~5–10%)

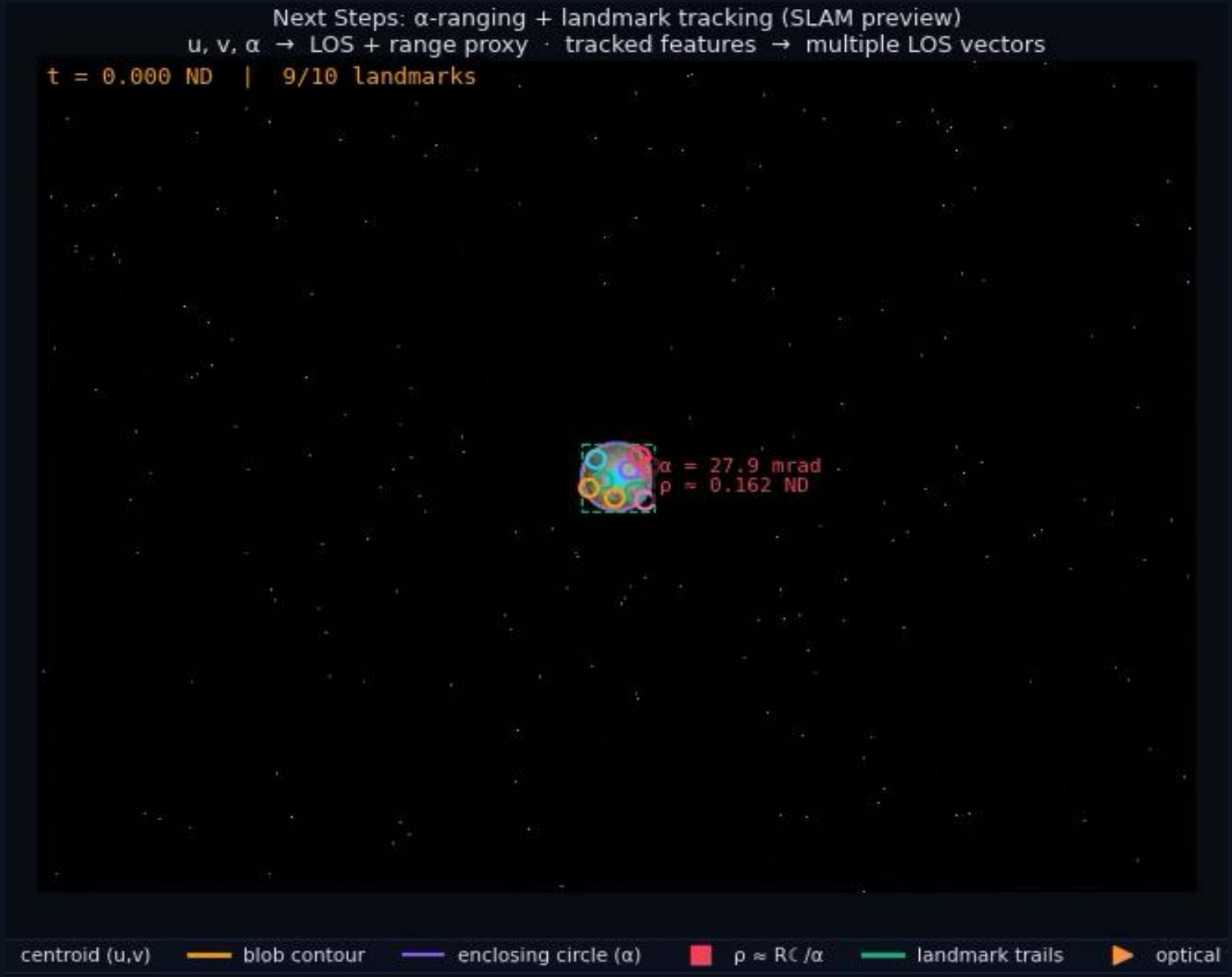


Robustness
Performance holds under: Noise, Dropout, Initial uncertainty

Future Work: Beyond Bearing-Only Navigation

α-ranging (range from vision)
Use angular size (α) → estimate range
Reduces scale ambiguity in bearing-only nav

Landmark tracking
Track multiple surface features over time
Generate multiple LOS vectors per frame
Improves observability & robustness



Towards SLAM-like navigation

Joint estimation of:
spacecraft state
feature geometry

Enables:
navigation in unknown environments
autonomy without prior maps

A path toward fully autonomous optical navigation in cislunar space

References and Resources

Core Texts

[1] Crassidis, J. L., & Junkins, J. L. Optimal Estimation of Dynamic Systems, 2nd ed., CRC Press, 2011

[2] Markley, F. L., & Crassidis, J. L. Fundamentals of Spacecraft Attitude Determination and Control, Springer, 2014

[3] Christian, J. A., & Robinson, S. B. Fundamentals of Spacecraft Optical Navigation, Wiley, 2022

[4] Hughes, P. C. Spacecraft Attitude Dynamics, Dover, 2004

Course / Academic Resources

UIUC AE 598 / AE 48: Spacecraft Navigation & Estimation

UIUC AE 598: Optical Navigation & Space Systems

NASA Goddard: Optical Navigation Techniques (public materials)

Methods & Theory

[5] Maybeck, P. S. Stochastic Models, Estimation, and Control, Vol. 1, Academic Press

[6] Bar-Shalom, Y., Li, X. R., & Kirubarajan, T. Estimation with Applications to Tracking and Navigation, Wiley

Project Code

GitHub Repository: https://github.com/varshakrishnakumar/cislunar-optical-nav